Canvas Course Manager

User Guide

Version 1.0

Harvard University

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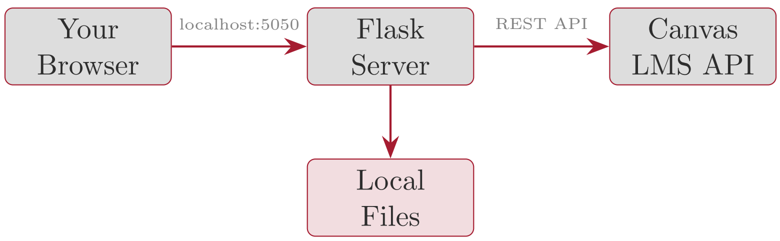
Getting Started

Overview

Canvas Course Manager is a locally-run web application that gives instructors a unified interface for managing Canvas LMS courses. Instead of navigating between multiple Canvas web pages, you get a single dashboard with tabs for browsing course content, managing files, editing assignments and quizzes, tracking student progress, and viewing a course calendar.



The Flask server acts as a proxy between your browser and the Canvas API, handling authentication and parallel data fetching to keep the UI responsive.

Installation

1. **Clone the repository:**

```
git clone https://github.com/astrostubbs/CanvasFrontEnd.git
cd CanvasFrontEnd
```

2. **Install Python dependencies:**

```
pip install -r requirements.txt
```

3. **Configure your Canvas API token** (see Section 1.3).

4. **Run the server:**

```
set -a; source .env; set +a
python canvas_gui.py
```

5. Open `http://localhost:5050` in your browser.

Obtaining a Canvas API Token

1. Log in to your Canvas instance (e.g., `canvas.instructure.com`).

2. Navigate to **Account** → **Settings**.

3. Scroll down to **Approved Integrations**.

4. Click + **New Access Token**.
5. Enter a purpose (e.g., "Course Manager") and optionally set an expiry date.
6. Click **Generate Token** and copy the token immediately.

Important

Your API token grants full access to your Canvas account. Never commit it to version control or share it. Store it in a `.env` file (which is git-ignored) or export it as an environment variable.

Environment Configuration

Copy the example configuration and fill in your values:

```
cp .env.example .env
```

Edit `.env` with your settings:

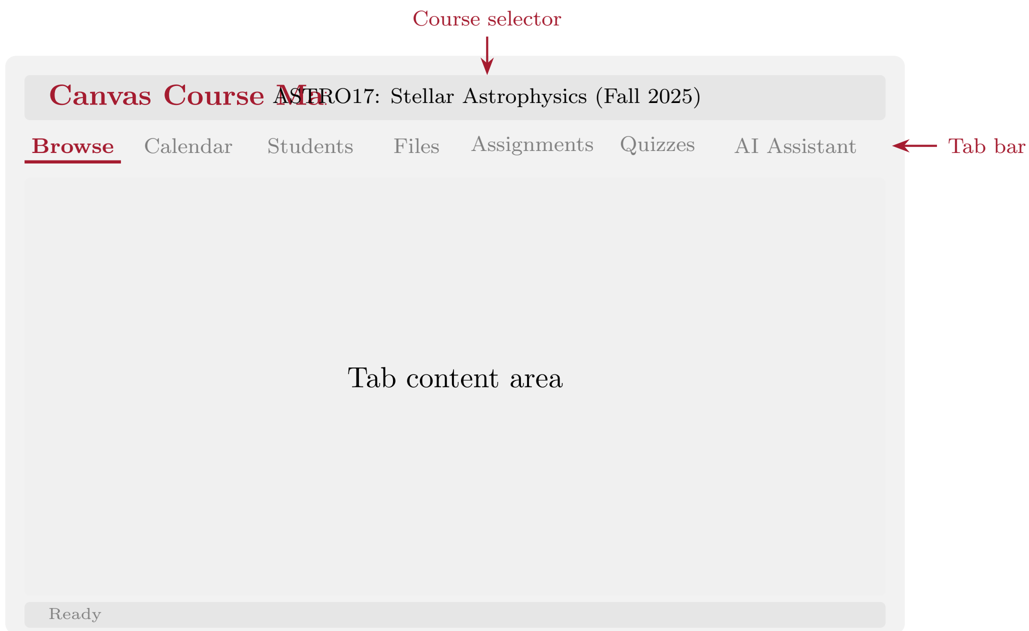
```
CANVAS_API_TOKEN=1234~AbCdEfGhIjKlMnOpQrStUvWxYz
CANVAS_API_URL=https://canvas.instructure.com/api/v1
LOCAL_ROOT=~/Documents/MyCourse
```

Table 1: Environment Variables

Variable	Required	Description
`CANVAS_API_TOKEN`	Yes	Your Canvas personal access token
`CANVAS_API_URL`	Yes	Canvas API base URL (with `/api/v1`)
`LOCAL_ROOT`	No	Root directory for local file browser

The Interface

When you first open the application, you see a header with a course selector and a row of seven tabs. Select a course from the dropdown to populate all tabs with that course's data.



The tabs are:

1. **Browse** — Course structure overview with expandable modules
2. **Calendar** — Month-by-month calendar of assignments and quizzes
3. **Students** — Student performance analytics and progress tracking
4. **Files** — Two-panel file manager (local + Canvas)
5. **Assignments** — Full-screen assignment editor
6. **Quizzes** — Full-screen quiz and question editor
7. **Page Editor** — Create and edit Canvas pages
8. **LaTeX → Canvas** — Convert LaTeX equations to Canvas format with live preview
9. **AI Assistant** — Chat interface powered by Claude

Browse Tab

The Browse tab shows a hierarchical view of your course content.

Global Links

At the top, quick-access links open the Canvas web interface for common pages: the course home, gradebook, syllabus, all files, and the people list.

Module Tree

Below the links, each module appears as an expandable row. Click a module to load and display its contents, **grouped by type**:

Module 2: Data Analysis 8 items

ASSIGNMENTS (3)
Problem Set 2: Regression
Lab: Linear Models
Homework 2 Revisions

QUIZZES (1)
? Module 2 Review Quiz

PAGES (2)
¶ Lecture Notes: OLS
¶ Reading: Chapter 4

FILES (2)
§ dataset.csv, notebook.ipynb

Tip

Click any item to open it directly in Canvas in a new browser tab. Items are lazy-loaded — the module contents are only fetched from Canvas when you click to expand the module.

Calendar Tab

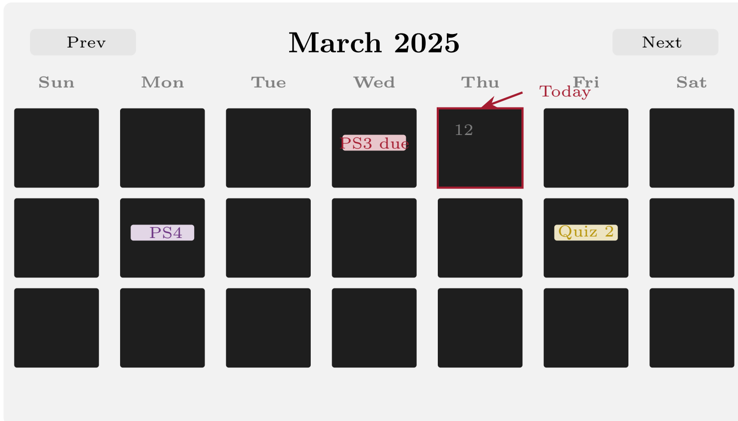
The Calendar tab provides a month-by-month view of your course schedule.

Navigation

Use the **Prev**/**Next** buttons to move between months, or click **Today** to jump to the current month.

Events

- **Assignment due dates** appear as red-tinted badges on their due date.
- **Availability windows** are shown as purple markers on the unlock date (when the assignment becomes available).
- **Quiz due dates** appear as amber/yellow badges.



Tip

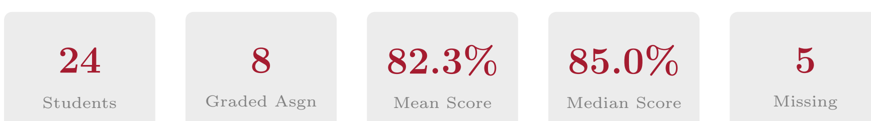
Click any event badge to open it directly in Canvas. Hover over a badge to see the full name and point value.

Student Progress Tab

The Students tab provides a comprehensive analytics dashboard for tracking student performance. This is designed to answer the questions instructors ask most often: Who is falling behind? What's the grade distribution? Which assignments are students struggling with?

Summary Statistics

Five summary cards appear at the top:



Ranked Student Scores

A horizontal bar chart shows every student's current percentage score, ranked from highest to lowest. Bars are color-coded:

- Green ($\geq$ 90%),
- Yellow (70–89%),
- Red ($<$ 70%).

This makes it easy to spot students who may need additional support.

Grade Distribution

A histogram-style bar chart groups students into letter-grade buckets: A (90–100), B (80–89), C (70–79), D (60–69), F (< 60). This gives you an at-a-glance view of the overall class performance distribution.

Missing Assignments

A sorted list of students who have missing submissions, ordered by the number of missing assignments (most first). Hover over a row to see which specific assignments are missing.

Important

The "missing" flag comes from Canvas. An assignment is marked missing if it is past due and the student has not submitted anything. If you use "on paper" submission types, Canvas may incorrectly flag those as missing.

Late Submissions

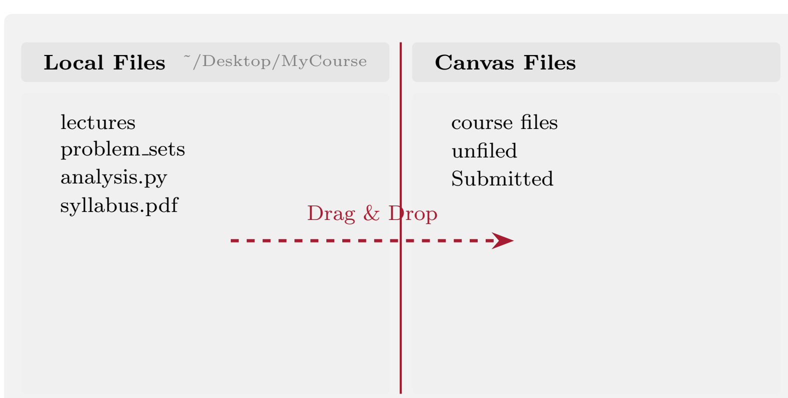
Similar to the missing list, this shows students with late submissions and the count of late work.

Per-Assignment Averages

A bar chart showing the class average score for each graded assignment. This helps identify assignments that were particularly challenging (low average) or straightforward (high average).

File Transfer

The Files tab provides a two-panel file manager: your local files on the left, Canvas course files on the right.



Uploading Files to Canvas

1. In the left panel, locate the file you want to upload.
2. **Drag** the file from the left panel and **drop** it onto a Canvas folder in the right panel.
3. A confirmation dialog appears letting you choose the destination folder.
4. Click **Upload** to transfer the file.

The upload uses Canvas's three-step upload process (notify, upload, confirm) and handles it automatically.

Browsing Canvas Files

Click any folder in the right panel to expand it and see its contents. Click a file to open/-download it in a new tab.

Tip

The local file panel starts at the directory specified by the `LOCAL_ROOT` environment variable. Set this to your course's working directory for quick access.

Assignment Editing

The Assignments tab provides a full-screen editor with a sidebar list and a main editing area.

Editing an Existing Assignment

1. Click an assignment in the sidebar to load it into the editor.
2. Edit any field: name, description, due date, availability window, points, grading type.
3. A green **Save Changes** button appears when you modify any field.
4. Click **Save Changes** to push the update to Canvas.

Creating a New Assignment

1. Click the + **New** button at the top of the sidebar.
2. Fill in the assignment name (required), description, due date, points, submission type, and grading type.
3. Click **Create Assignment**. It will be created as a draft (unpublished).
4. Publish it when ready using the **Publish** button in the editor.

Publishing and Deleting

- **Publish/Unpublish:** Toggle the publication state directly from the editor.
- **Delete:** Click the red **Delete** button. A confirmation dialog will appear.
- **Open in Canvas:** A link button opens the assignment in the Canvas web UI.

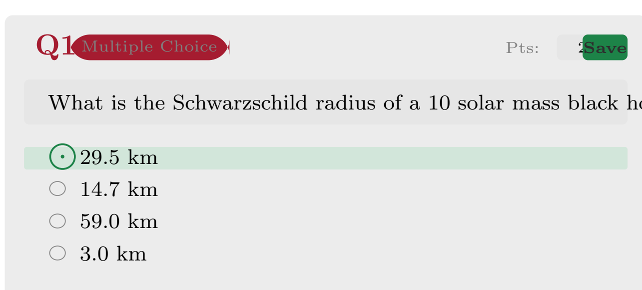
Quiz Editing

The Quizzes tab has the same sidebar-plus-editor layout. Selecting a quiz loads all its questions as editable cards.

Editing Questions

Each question card shows:

- The question number and type (Multiple Choice, True/False, Essay, etc.)
- An editable text area for the question text (supports HTML)
- An inline points input
- For multiple-choice questions: radio buttons to select the correct answer, with editable answer text fields



Workflow: Edit and Save

1. Modify the question text, points, or answer text.
2. For multiple-choice: click the radio button next to the new correct answer.
3. The card border turns red and a **Save** button appears.
4. Click **Save** to update the question on Canvas.

Note

Published quizzes *can* be edited via the API as long as no student has started a submission attempt. If students have already taken the quiz, Canvas may restrict certain changes.

Adding Questions

Click **+ Add Question** to expand a form where you can:

- Choose the question type (Multiple Choice, True/False, Short Answer, Essay, Numerical)
- Enter the question text
- Set point value
- Add answer options and select the correct one
- Click **Add Question** to save it to the quiz

Creating a New Quiz

Click **+ New** in the sidebar to open the quiz creation form. Set the title, description, quiz type (graded, practice, survey), and optional time limit. New quizzes are created as drafts — publish when ready.

LaTeX to Canvas Converter

The **LaTeX → Canvas** tab solves one of the most frustrating problems in Canvas: getting mathematical equations to display correctly. Canvas does not natively support LaTeX input, but it does render equations via specially encoded image URLs. This tab automates the conversion.

How It Works

The tab provides a side-by-side editor:

- **Left panel:** An HTML editor where you write your page content.
- **Right panel:** A live preview showing how Canvas will render it.

Canvas equations use a special URL format:

```
https://canvas.instructure.com/equation_images/<encoded-latex>
```

where the LaTeX is double-URL-encoded. This tab handles the encoding automatically.

Inserting Equations

1. Type LaTeX in the **equation bar** (e.g., `\frac{dN}{dM} = kM^{-\alpha}`).
2. A live preview appears showing how the equation will render.
3. Click **Insert Equation** to insert the Canvas-formatted `<img>` tag at the cursor position in the editor.
4. The right panel updates to show the rendered result.

Quick-Insert Symbols

A toolbar provides one-click buttons for common LaTeX symbols: fractions, square roots, integrals, summations, Greek letters ($\alpha, \beta, \gamma, \lambda$), solar mass ($M_\odot$), operators ($\pm, \approx, \leq, \geq, \partial, \nabla$), and more.

Batch Conversion

If you prefer to write content using `$...$` (inline) and `$$...$$` (display) delimiters, click **Convert All $...$** to automatically find all LaTeX delimiters in the editor and replace them with Canvas equation image tags.

Tip

This is the fastest workflow: write your page content using familiar LaTeX delimiters, then click one button to convert everything to Canvas format.

AI-Assisted LaTeX

Click **AI Assist** and describe the equation you need in plain English (e.g., "Stefan-Boltzmann law" or "integral of x squared from 0 to infinity"). Claude will generate the LaTeX code and populate the equation bar for you.

Loading and Saving Canvas Pages

- **Load from Canvas Page:** Fetches an existing page's HTML content into the editor for editing.
- **Save to Canvas Page:** Pushes the editor content directly to a Canvas page.
- **Copy HTML:** Copies the raw HTML to your clipboard for pasting elsewhere.

AI Assistant

The AI Assistant tab provides a chat interface powered by Claude Code. It can interact with your Canvas course through MCP (Model Context Protocol) tools.

What It Can Do

- Generate quiz questions on a topic
- Write assignment descriptions and rubrics
- Draft course announcements
- Look up student enrollment and course data
- Answer questions about Canvas course content

Using the Chat

1. Select a course from the dropdown (gives the AI context about which course to query).
2. Type your request in the text box or click a suggestion chip.
3. Press Enter or click Send.
4. The response streams in real-time.

MCP Setup

For the AI to access Canvas data directly, you need a Canvas MCP server configured. Create a `canvas.mcp.json` file in the project directory:

```
{
    "type": "http",
    "url": "https://your-mcp-server/mcp",
    "headers": {
        "X-Canvas-API-Token": "your_token",
        "X-Canvas-API-URL": "https://your-canvas.instructure.com"
    }
}
```

Important

The `canvas.mcp.json` file contains your API token and is excluded from version control by `.gitignore`. Never commit this file.

Common Workflows

Weekly Check-In: Are Students On Track?

1. Open the **Students** tab.
2. Check the **Missing Assignments** card — follow up with students who have multiple missing items.
3. Review the **Grade Distribution** — if it's skewed low, consider reviewing the material.
4. Look at **Per-Assignment Averages** — a low average on a specific assignment may indicate unclear instructions.

Preparing Next Week's Content

1. Open the **Calendar** tab to see what's due and when.
2. Switch to **Assignments** to create or edit upcoming assignments. Set due dates and availability windows.
3. Use the **Files** tab to upload any new materials (lecture slides, datasets, readings) by dragging from local to Canvas.
4. Switch to **Quizzes** to create a review quiz. Use **AI Assistant** to generate question ideas.

Creating a Quiz from Scratch

1. Go to the **Quizzes** tab and click **+ New**.
2. Enter a title and set the quiz type (Graded Quiz, Practice Quiz, etc.).
3. Click **Create Quiz**.
4. Click **+ Add Question** and fill in the question details.
5. Repeat for each question.
6. When ready, click **Publish** to make it available to students.

Uploading Course Materials

1. Go to the **Files** tab.
2. In the left panel, navigate to the file you want to upload.
3. Drag the file onto the target Canvas folder in the right panel.
4. Confirm the upload in the dialog.

Troubleshooting

Problem	**Solution**
"API error: 401"	Your Canvas API token is invalid or expired. Generate a new one and update `.env`.
No courses appear in dropdown	Verify `CANVAS_API_URL` is correct and includes `/api/v1`. Ensure your token has teacher enrollment.
AI Assistant: "Claude CLI not found"	Install Claude Code: `npm install -g @anthropic-ai/claude-code`
AI Assistant: permission errors	Ensure `canvas.mcp.json` exists in the project directory with correct credentials.
Port 5050 already in use	Another instance is running. Kill it: `lsof -ti:5050 | xargs kill -9`
Student Progress shows no data	The course may not have graded assignments yet, or students haven't submitted work.

Security Considerations

- The application runs **locally only** on `127.0.0.1:5050`. It is not designed for public deployment.
- Your Canvas API token is never sent to any service other than your Canvas instance.
- The `.gitignore` excludes `.env` and `canvas.mcp.json` to prevent accidental token commits.
- The AI Assistant inherits your shell environment. Ensure your Anthropic/Bedrock API keys are set via environment variables, not hardcoded.

- If you share this tool with others, remind them to generate their own API tokens.